

Which language, and what that admits

A signal is read; a convention is declared; nothing is inferred. The whole design of this layer is that distinction.

1 • resolving the language

declared outranks detected; the script only adds

1. declared

language on /v1/search
A deliberate statement about a corpus, made at READ time — so a vault ingested under one language answers correctly under another with no re-ingest.
Outranks everything below.

2. else the drawer itself

language_of_drawer
A text carrying der, die, und, nicht is German. Only CLOSED-class words vote — a loanword does not get one.
Decisive or nothing: three hits and twice the runner-up, else stay silent.

+ the query's script, added

morph_lang_by_script(query)
Greek · Georgian · Hangul — one script, one language. Cyrillic → Russian and Devanagari → Hindi, labelled approximate.
Joins as a SECOND inflecting language; never replaces 1 or 2, nor feeds suffixes.

Read per candidate, not per query: a vault may hold several languages, and the drawer is the unit that has one.

2 • five rules, every one of them pairwise

each answers about two strings and builds no class

suffix_family
SHAPE, not length.
-er is German-only:
Kind/Kinder, Haus/Häuser

inflection_family
A CLOSED set of endings
per language, stem ≥ 3:
libro/libri, книга/книги

agglutinative_family
Endings that stack:
Turkish, Korean,
Georgian, Hungarian

ar_root_family
144 roots × 20 patterns
= 2859 forms. A pair
meets only via ONE root.

IRREGULAR
201 pairs no rule
over shape reaches:
go/went, gehen/ging

Why not a stemmer

A stemmer builds an equivalence class, and one false friend poisons every member. Measured: Snowball Greek merges πολύ (much) with πόλη (city). A pairwise rule costs only its own pair.

Why Arabic needed its own table

An ALLOWLIST: a form the table cannot generate matches nothing. Three subsequence rules failed: بيت→بيوت and يجيب→يجب are the same string operation; no rule over shape separates them.

3 • three channels, and only two of them are a claim about words

lexical_exact
The drawer said the word.
admits

lexical_morph
It holds a word BUILT ON it.
admits — and says so separately

semantic
Cosine, above the gate the
EMBEDDER itself calibrates.

Measured — 191 paradigm pairs, 19 languages

lexical recall, declared	55.0% → 100%
lexical recall, undeclared	→ 100%
pairs resting on the embedder alone	20 → 0
negative controls	0 → 48

The prices, pinned rather than hidden

Declaring German is what makes -er legal, and -er also relates flow/flower, corn/corner, butt/butter. That is the trade, stated.
A recall measurement cannot justify a precision decision: adding -er moved the population figure by 0.21 links per query — invisible — and broke five negative controls. The controls are what caught it.